



GlucoTwin

Two-hour glucose event research

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Synthetic demo. Not for clinical decisions.

The review question

What changed, and can this forecast be trusted?
A proposed clinician-facing workflow for simulated
Type 2 diabetes review.
Inspect inputs, advance replay, then compare the
observed outcome.

No clinician interview or usability study has been performed.

120 minutes ahead

Two consecutive future readings >180 mg/dL

15-minute samples. Both readings must lie in the next 120 minutes.

Current >180 : no new-event estimate. Missing future data: unknown outcome.
Two readings span 15 minutes, not 30.

Synthetic demo and real evidence

Browser demonstration

240 fictional people. Four selected held-out traces. Static context affects the generator by design.

Engineering evidence only.

Retrospective benchmark

100 adults, 109 sessions. Separate models and aggregate results. Age and recorded gender only.

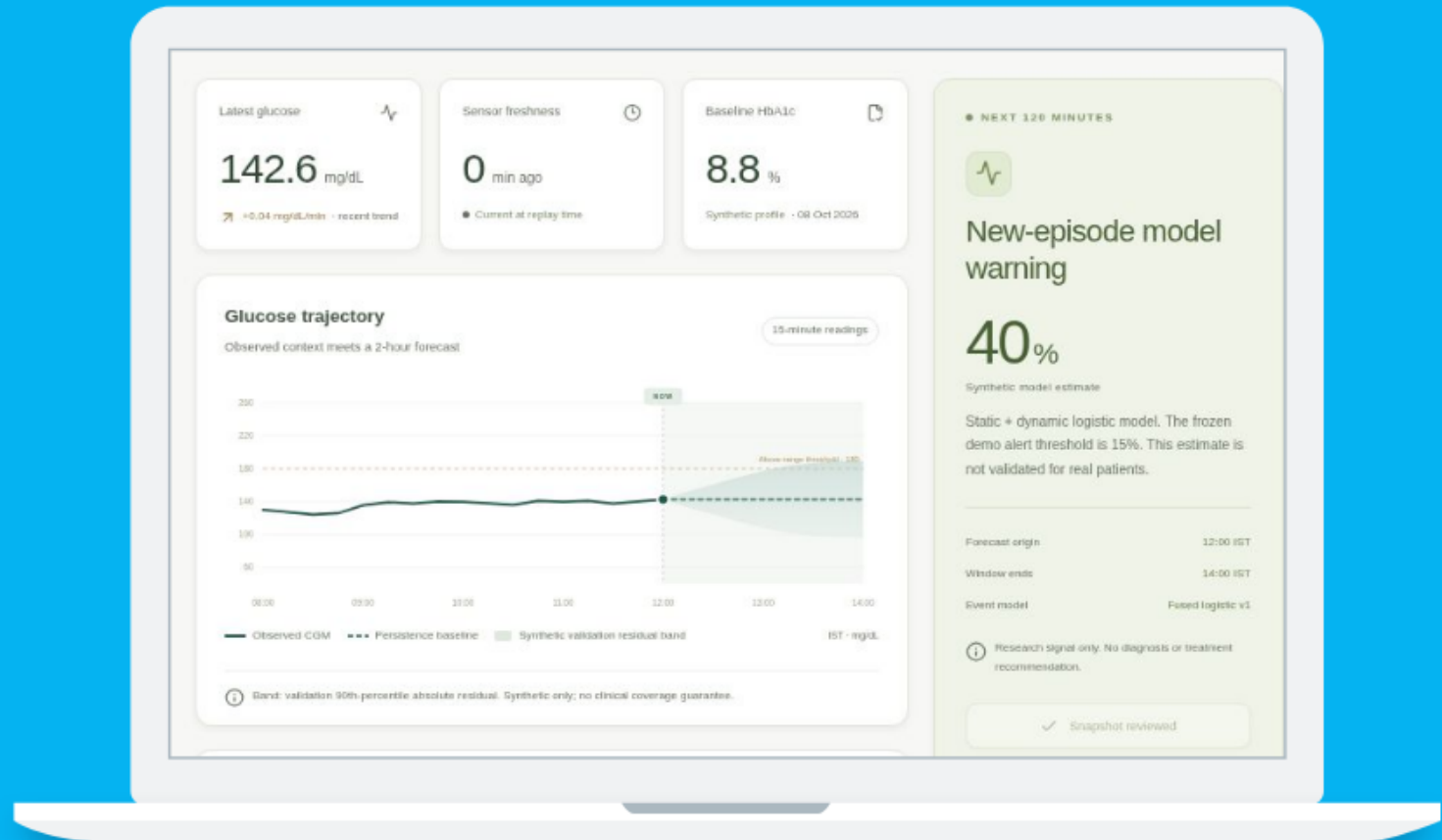
Baseline timing is assumed. All undated labs are excluded.

Working review loop

Select a fictional patient.

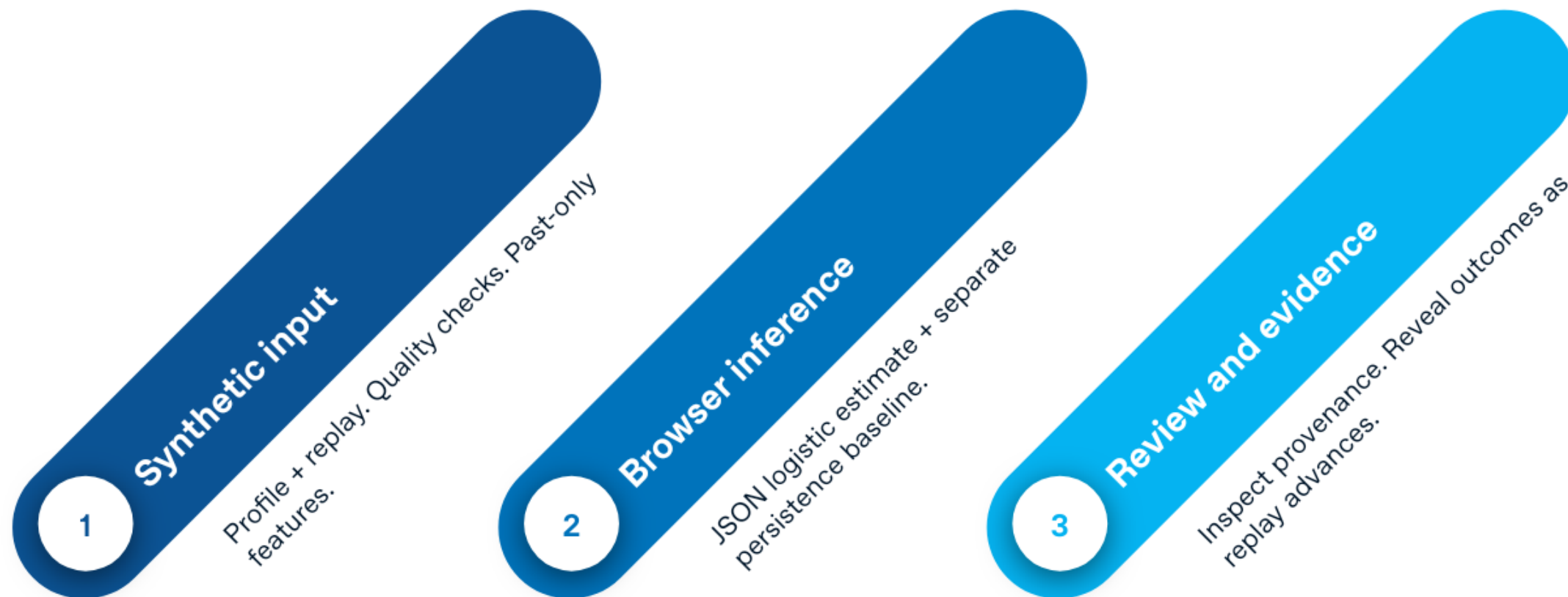
Advance replay.

Inspect inputs and evidence.



Browser-only inference. Session-only review notes. No clinical actions.

Implemented architecture



Offline Python training exports synthetic JSON. A separate local pipeline produces real aggregate evidence.

The outputs have different jobs

Event estimate

Logistic regression uses last-hour glucose statistics plus synthetic age, diabetes duration and pre-existing HbA1c.

A synthetic estimate, not clinical risk.

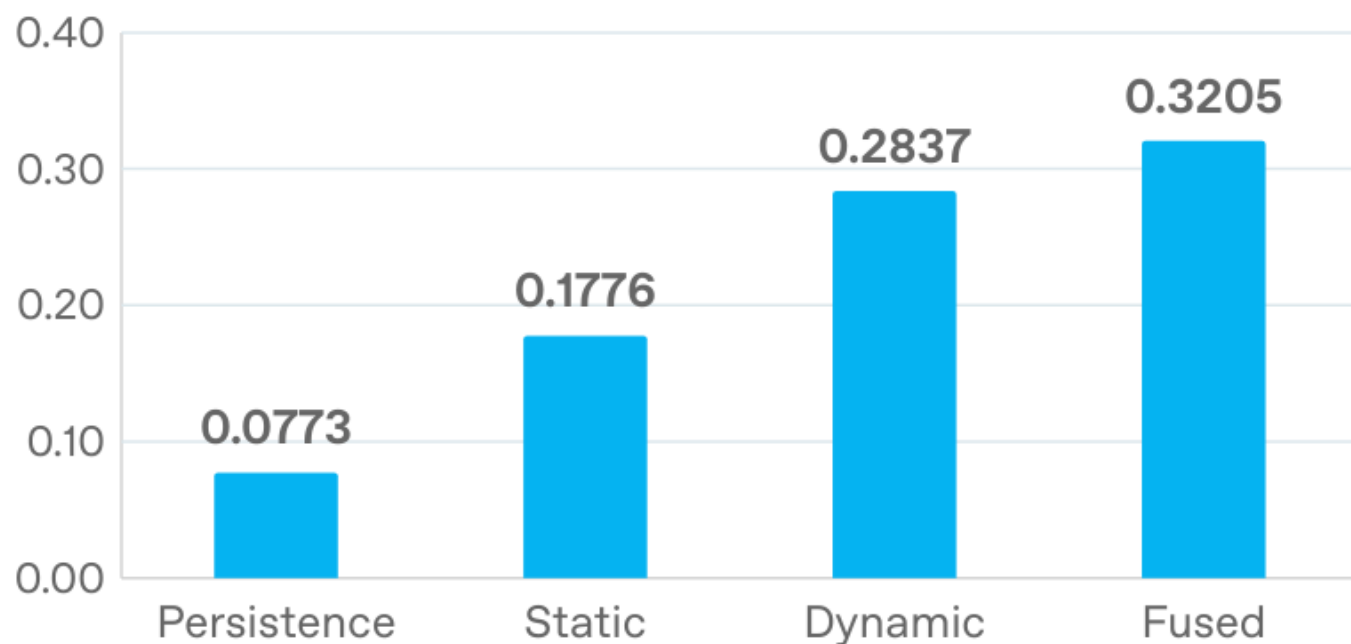
Glucose chart

Persistence holds current glucose constant. Bands use validation absolute residuals.

Separate models can disagree. Bands do not guarantee individual clinical coverage.

Synthetic held-out results

PR-AUC (average precision)



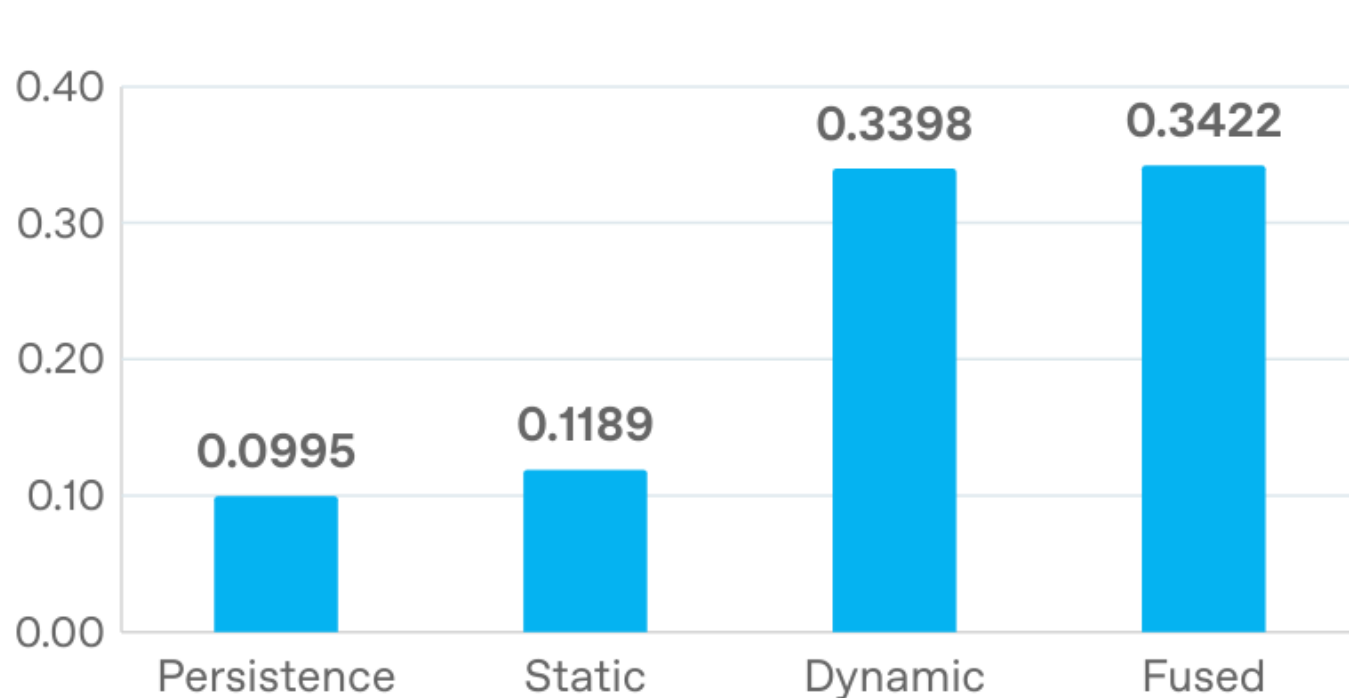
2.81
false alerts /
patient-day

48 test people / 30,248 windows. Toy-generator results only.

293/334 episodes warned. Median lead 30 min. Window precision 27.2%.

Real fusion gain is uncertain

PR-AUC (average precision)



+0.0024

PR-AUC difference

Paired 95% interval

-0.00248 to +0.00758

Includes zero

20 test people / 20,687 windows. A small retrospective feasibility study.

No meaningful fusion benefit or fitness for care established.

Real benchmark at the validation-selected 0.20 threshold

2.97

false alerts per recorded patient-day

302/326 episodes warned / Median lead: 45 minutes
754 false alerts over ~254 days / Window precision: 28.9%

Recorded-span days include gaps and current-high periods. Alerts use a 120-minute cooldown.

Failure cases and next validation



Selected illustrations

False positive and genuinely missed synthetic episode.



Data quality

Stale, invalid, disconnected and cold-start states withhold forecasts.



Evidence limits

No prospective study or Indian-population validation.



Human review

No clinician usability study or treatment workflow.



Privacy boundary

Real participant records stay outside the app and submission.



Next research step

External validation, verified input timing and review of alert burden.

Sources and contribution disclosure

Original code: MIT. Data and template licenses remain separate.

AI assistance: design, code, evaluation, tests and documentation.

Template: SlidesCarnival, Dercetus / Minimal Medical. CC BY 4.0.

Adapted layouts, text, charts and screenshot. slidescarnival.com

Original template credits: Unsplash. Stock photos are not used here.

ShanghaiT2DM: Zhao et al. (2023), CC BY 4.0.

[doi:10.1038/s41597-023-01940-7](https://doi.org/10.1038/s41597-023-01940-7)

Dataset v5: [doi:10.6084/m9.figshare.21600933.v5](https://doi.org/10.6084/m9.figshare.21600933.v5)

Modified into trailing features and aggregate results.